CytoBridge: A Calibrated, Any-Modality Framework for Cervical Screening Support

A B.Sc. Thesis

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Abstract

Cervical cancer is largely preventable, but screening relies on scarce expert cytologists. Two data sources can support screening, and they carry *different, complementary* signals: a Pap-smear image reports *cytological abnormality*, whereas a patient questionnaire supports a *high-risk HPV infection-risk* estimate. These sources are almost never collected for the same patient, so a jointly trained multimodal model cannot be built from paired data. We study whether two *independently* trained experts can instead be combined at the decision level. We build a cytology expert (an EfficientNet-B0 distilled from an EfficientNet-B4 teacher) and an HPV-risk expert (a RandomForest model on $n = 12{,}634$ women from NHANES 2003–2016), and combine their calibrated probabilities with three components: CO-PRIOR, a prior-composition step that re-anchors *each* expert to the deployment prevalence *before* pooling; an asymmetric combiner; and split-conformal abstention. On its held-out validation split the cytology expert reaches 0.966 accuracy / 0.998 AUROC (five-class); the HPV-risk expert reaches 0.721 accuracy / 0.688 AUROC under cross-validation. The purpose of the fusion is *not* to raise discrimination beyond the stronger single expert—it does not—but to add complementary HPV-risk information, keep the fused probability calibrated when the local prevalence differs from training, and operate when either modality is missing. We report each expert's results and the fusion behaviour, and we discuss the domain-shift and patient-pairing limitations candidly.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence**; • **Applied computing** → **Health informatics**.

Keywords: cervical screening support, knowledge distillation, calibration, decision-level fusion, HPV risk, conformal prediction

1 Introduction

1.1 Background for the non-specialist

What this work does, and does not, predict

Nearly all cervical cancer is caused by persistent high-risk *human papillomavirus* (HPV) infection [15]. This work supports *screening*, not diagnosis. It builds two models that predict *different* endpoints: (i) a cytology model that classifies cervical-cell *morphology* (later reduced to abnormal/normal), and (ii) a questionnaire model that estimates *high-risk HPV positivity*. Neither predicts cervical cancer directly; they are complementary screening signals.

Two independent lines of evidence

The two models are best seen as two *independently collected* lines of evidence about the same clinical question—one morphological, one behavioural/epidemiological. They do not observe the same patient, and they answer related but distinct questions, so combining them is a problem of pooling heterogeneous evidence rather than of averaging two opinions on one measurement.

Cervical cancer remains a leading cause of cancer death in women despite being preventable [13], and expert cytologists are scarce, motivating machine-learning support for both the image read and the risk estimate.

1.2 The problem, and why paired data are unavailable

A standard multimodal model is trained on records that contain *all* modalities for *each* patient. That option is not available here: public cervical-cytology image datasets and public HPV-risk questionnaire datasets are collected in different studies and share no patients, and there is no key linking a cell image to a questionnaire respondent. A jointly trained image+questionnaire model therefore cannot be built or evaluated on real paired data. The practical question this work targets is consequently:

Gaps this work addresses

G1 — deployable, explainable cytology. High-accuracy cervical-cytology classifiers are typically built on large convolutional backbones [10, 14], which strains modest hardware and gives little insight into *why* a cell is flagged unless paired with post-hoc explanation [12]. *Addressed by* a distilled, single-GPU EfficientNet-B0 expert with calibration and Grad-CAM attribution (§3.2, §5.1).

G2 — a calibrated HPV-risk estimate. Questionnaire-based HPV-risk models are commonly reported by accuracy or AUROC alone, without a calibrated probability or a stated operating point [5], yet a screening flag is only actionable as a well-calibrated probability. *Addressed by* an HPV-risk expert with isotonic calibration and full per-model reporting (§3.3, §5.2).

G3 — combining evidence without paired data. With no shared patients and each expert calibrated to its own base rate, classical opinion pooling [8] cannot simply average the two outputs; the priors must be reconciled [11] and uncertain cases handled with a coverage guarantee [2]. *Addressed by* CO-PRIOR re-anchoring, an asymmetric combiner, and conformal abstention (§3.4, §5.3).

1.3 Research question and hypothesis

RQ. Can independently trained cytology and questionnaire HPV-risk models be combined at the decision level to give a calibrated, any-modality screening-support output when patient-level multimodal pairing is unavailable? **Hypothesis.** Re-anchoring each expert's prior to the deployment prevalence before an asymmetric pooling step can combine the two complementary signals and keep the fused probability calibrated across prevalences, without requiring paired training data—while not necessarily improving discrimination over the stronger single expert.

1.4 Contributions and implementation choices

We separate what is a contribution from what is a standard implementation choice. *Contributions:*

- **A decision-level framework for unpaired complementary modalities** that combines two independently trained experts without any patient-level pairing (§3.4).
- **CO-PRIOR**: prior correction applied *independently to each heterogeneous expert before* decision-level pooling, so both outputs live on one deployment-prevalence scale (§3.4).
- **A calibrated, any-modality operating design**: prevalence re-anchoring, an asymmetric cascade, and split-conformal abstention, evaluated for calibration and coverage (§5.3).

Implementation choices (standard components, reported for completeness): EfficientNet-B4→B0 knowledge distillation for the cytology expert, and a class-weighted RandomForest ensemble for the HPV-risk expert.

2 Related Work

Deep cytology classification. Convolutional models on single-cell benchmarks such as SIPaKMeD [10] and the Herlev Pap-smear set [6] report high in-distribution accuracy; EfficientNet backbones [14] with knowledge distillation [4] yield a compact model, the recipe we adopt for the cytology expert.

Questionnaire HPV-risk models. Models predicting HPV or cervical outcomes from questionnaire and demographic variables typically use tree ensembles with class-imbalance handling [5]; we follow this recipe for the HPV-risk expert, using class weights rather than synthetic oversampling [3], and add explicit probability calibration.

Decision-level fusion and calibration. Combining independent classifiers by opinion pooling and the sum/product rules is classical [8]. Adjusting a classifier's outputs to a new prior is a single-classifier procedure [11]; CO-PRIOR applies it *per expert before* pooling heterogeneous, differently-calibrated models. Split-conformal prediction [2] supplies the abstention machinery with a coverage guarantee.

3 Methodology

3.1 Problem formulation and notation

Let x_{img} be a cervical-cell image and x_{tab} a questionnaire feature vector. The two frozen experts produce calibrated posteriors

$$p_{\text{img}} = f_{\text{img}}(x_{\text{img}}) \in [0,1], \qquad p_{\text{tab}} = f_{\text{tab}}(x_{\text{tab}}) \in [0,1], \tag{1}$$

where p_{img} is the probability of cytological abnormality and p_{tab} is the probability of high-risk HPV positivity. Each expert was calibrated at its own training prevalence $(\pi_{\text{img}}, \pi_{\text{tab}})$; the deployment prevalence τ is declared. Either input may be absent. The fused output is

$$p_{\text{fused}} = F\big(p_{\text{img}}, p_{\text{tab}}, \tau\big), \tag{2}$$

and the system returns p_{fused} with a decision in $\{\text{RELEASE}, \text{DEFER}, \text{FLAG}\}$. *Reading Eq. (1)–(2):* the two experts are trained and calibrated separately on disjoint data; F is the decision-level operator (§3.4) that reconciles their priors to τ and pools them. Figure 1 shows the end-to-end system.

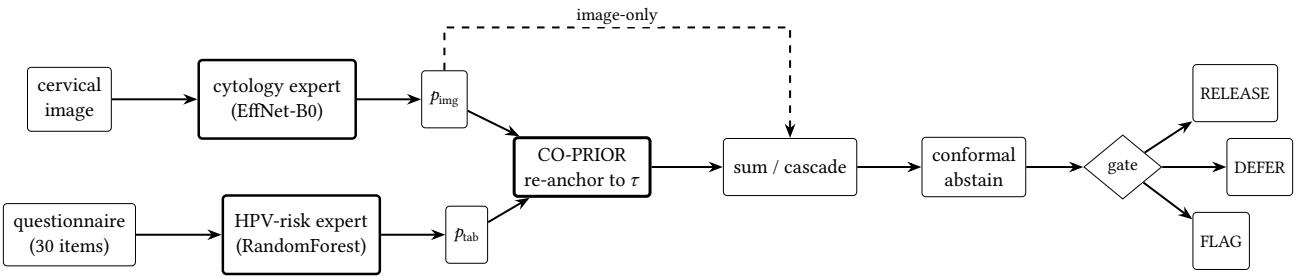


Figure 1. End-to-end system. Two independently trained experts predict different endpoints (cytological abnormality; high-risk HPV positivity); CO-PRIOR re-anchors each to the deployment prevalence τ before the combiner; a conformal check and gate produce RELEASE/DEFER/FLAG. A missing modality bypasses the combiner.

3.2 Cytology expert

Dataset. SIPaKMeD [10] contains 5,015 Pap-smear images (4,049 isolated single cells and 966 cluster crops) in five morphology classes: Dyskeratotic, Koilocytotic, Metaplastic, Parabasal, Superficial–Intermediate. The classes are near-balanced (largest-to-smallest ratio 1.19). The first two are the abnormal classes; koilocytosis is a morphological correlate of HPV infection.

Pre-processing. Images are resized to 380×380. Training images receive random horizontal and vertical flips ($p = 0.5$ each), random rotation ($\pm 72^{\circ}$), and scale jitter (0.9–1.1×), then ImageNet normalisation; validation images are only resized and normalised.

Knowledge distillation. An EfficientNet-B4 teacher [14] is distilled into an EfficientNet-B0 student [4]. Schedule: teacher stage 1 (backbone frozen, head only, 30 epochs, lr 10^{-3}); teacher stage 2 (full fine-tune, 50 epochs, lr 10^{-5}, `ReduceLROnPlateau` factor 0.3, early stopping); student (40 epochs, lr 5×10^{-5}, dropout 0.3, temperature $T = 4.0$, $\alpha = 0.1$), batch size 32, BF16. The distillation loss is

$$\mathcal{L} = \alpha\, \mathrm{CE}\big(y, \sigma(z_s)\big) + (1-\alpha)\, T^2\, \mathrm{KL}\Big(\sigma\big(\tfrac{z_t}{T}\big) \,\Big\|\, \sigma\big(\tfrac{z_s}{T}\big)\Big). \quad (3)$$

Reading Eq. (3): z_s, z_t are student/teacher logits, σ the softmax. The first term is cross-entropy against the hard label y; the second is the KL divergence between the teacher's and student's temperature-softened distributions, transferring inter-class similarity; the T^2 factor matches the two gradient scales. At inference the five-class softmax is reduced to a single abnormal probability $p_{\text{img}} = p(\text{Koilocytotic}) + p(\text{Dyskeratotic})$. Figure 2 summarises the pipeline.

3.3 HPV-risk expert

Dataset and features. Seven NHANES cycles (2003–2016) give $n = 12{,}634$ women aged 18–59 with a valid self-collected HPV swab, merged on the respondent identifier across the demographic, sexual-behaviour, reproductive, smoking, and immunisation questionnaires. The label is high-risk HPV positivity (any of 14 oncogenic genotypes on the Roche Linear Array), present in 24.6% of the cohort. The features are 30 questionnaire items plus engineered exposure ratios (e.g. lifetime partners per sexually-active year).

Model. Missing values are median-imputed; class imbalance is handled by class weights rather than synthetic oversampling [3], so every model is trained on the real 12,634 records; the data are modelled with a RandomForest ensemble (600 trees, balanced class weights) and probabilities are isotonic-calibrated. Figure 3 summarises the pipeline; model selection is detailed in §4.

3.4 Decision-level fusion

Step 1 — CO-PRIOR.. Each expert is calibrated to its own base rate, so before pooling we strip that prior and install the deployment prevalence τ:

$$\operatorname{logit} \hat{p} \;=\; \operatorname{logit} p \;-\; \operatorname{logit} \pi \;+\; \operatorname{logit} \tau. \quad (4)$$

Reading Eq. (4): p is a raw posterior, π its training prevalence, τ the deployment prevalence; subtracting $\operatorname{logit} \pi$ removes the trained-in base rate and adding $\operatorname{logit} \tau$ installs the deployment one. The single-classifier form of this adjustment is due to Saerens et al. [11]; the contribution here is applying it *independently to each heterogeneous expert before* decision-level pooling, so that two models with different priors become comparable.

Step 2 — combine. We pool the re-anchored posteriors with the Kittler sum rule [8], $p_{\text{sum}} = \frac{1}{2}(\hat{p}_{\text{img}} + \hat{p}_{\text{tab}})$.

Step 3 — asymmetric cascade (engineering safety heuristic). As a design choice—not a clinically validated rule—the combiner does not let a low questionnaire score reduce a high cytology score: $p_{\text{fused}} = \max(p_{\text{sum}}, \hat{p}_{\text{img}})$ when $\hat{p}_{\text{img}} \geq \theta$, else p_{sum}. This prevents the questionnaire from suppressing a positive cytological signal above the threshold θ.

Step 4 — abstain. Each expert carries split-conformal thresholds [2] calibrated on its own data; ambiguous or

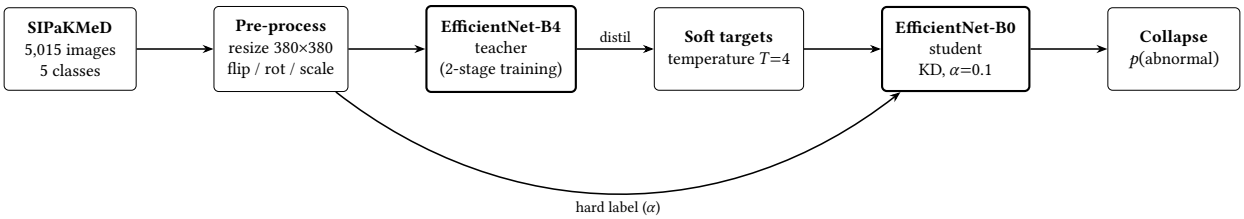


Figure 2. Cytology-expert training pipeline (left to right). Bold-bordered boxes are the trained networks: a two-stage B4 teacher distils a B0 student ($T = 4$, $\alpha = 0.1$) whose five-class softmax is reduced to p(abnormal).

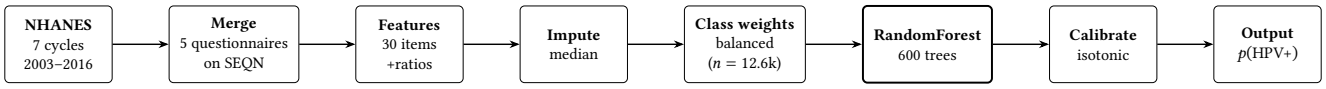


Figure 3. HPV-risk-expert training pipeline (left to right). Seven NHANES cycles are merged into a 30-feature table, imputed, class-weighted to offset imbalance, modelled with RandomForest, and isotonic-calibrated to p(HPV+). The bold-bordered box is the trained model.

lone-weak cases return DEFER, substantial disagreement returns FLAG, otherwise RELEASE.

4 Experimental setup

Data splits. The cytology expert uses a stratified 80/20 train/validation split of SIPaKMeD (seed 42); all reported cytology numbers are on this *validation* split (there is no separate held-out test set—see §7). The HPV-risk expert is evaluated by stratified 5-fold cross-validation on NHANES. Cross-dataset transfer uses Herlev, APACC, and HiCervix as external image sets.

Metrics. We report accuracy, balanced accuracy, macro-AUROC, F1, precision, recall, and, for probability quality, expected calibration error (ECE) with reliability diagrams. Discrimination (AUROC) is threshold-free; the operating-point metrics use a 0.5 threshold unless stated.

Model selection. For the HPV-risk expert we compared ExtraTrees, RandomForest, XGBoost, LightGBM [7], and k-NN under the same 5-fold protocol; gradient-boosted families were tuned with Optuna [1] (tree-structured Parzen sampler, objective = CV AUROC). RandomForest (600 trees, balanced class weights) gave the highest CV AUROC and is used throughout; its probabilities are isotonic-calibrated.

Fusion evaluation. Because no patient has both modalities, the fusion is analysed per expert (calibration, conformal coverage) and on a deployment cohort constructed from the experts' real per-class score distributions to study prevalence re-anchoring and combiner behaviour.

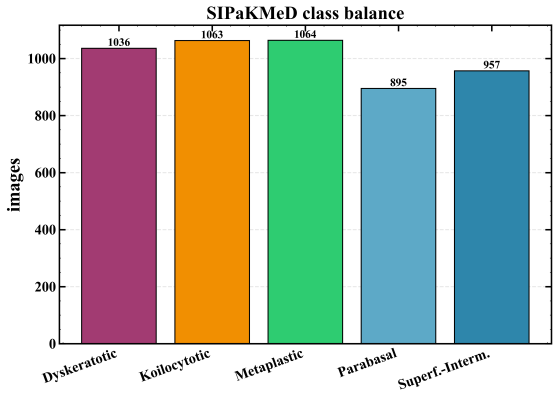


Figure 4. SIPaKMeD images per morphology class (imbalance ratio 1.19).

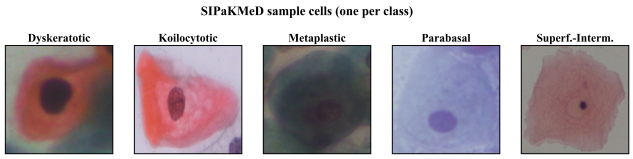


Figure 5. One representative cell per SIPaKMeD class.

5 Results

Results are grouped by component: the cytology expert (§5.1), the HPV-risk expert (§5.2), and the fused system (§5.3).

5.1 Cytology expert

Dataset. The five morphology classes are near-balanced (imbalance ratio 1.19; Figure 4), so no class dominates training; Figure 5 shows one cell per class.

Table 1. Cytology expert: five-class validation-split performance and external transfer (student).

Setting	Acc	Bal. acc	AUROC	ECE
Teacher B4 (SIPaKMeD val)	0.943	0.946	0.995	0.013
Student B0 (SIPaKMeD val)	**0.966**	**0.968**	**0.998**	0.016
Student, Herlev [6]	—	0.651	0.728	—
Student, APACC	—	0.535	—	—
Student, HiCervix	—	0.562	—	—

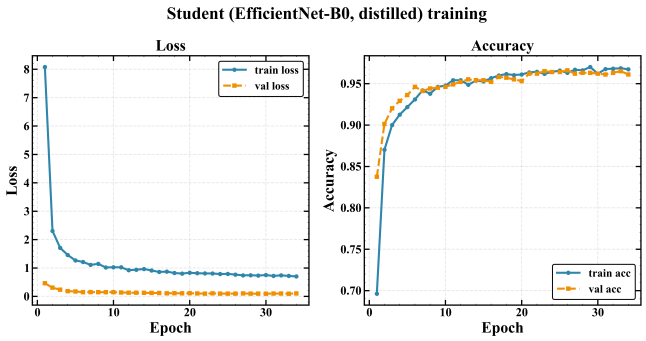


Figure 6. Student training/validation loss and accuracy across distillation epochs.

Classification performance. On the validation split the distilled student attains 0.966 accuracy, 0.968 balanced accuracy, and 0.998 macro-AUROC (Table 1), exceeding the larger teacher despite having roughly a quarter of the parameters (∼ 4M vs ∼ 17.5M). Balanced accuracy equal to accuracy indicates the model is not exploiting a majority class. Training and validation curves (Figure 6) converge without a growing train/validation gap. Because these are validation-split numbers, they should be read as in-distribution performance, not as an independent test estimate.

Errors, discrimination, calibration. The confusion matrix (Figure 7) is diagonal, with residual confusions between morphologically adjacent classes. The per-class ROC curves and the reliability diagram (Figure 8) give ECE = 0.016, indicating close agreement between predicted probabilities and empirical frequencies on the validation split.

Attribution. Grad-CAM [12] for the abnormal-class logit indicates that attribution concentrates on nuclear and perinuclear regions (Figure 9); this localises the gradient signal to morphologically relevant regions but does not by itself establish that the model uses the underlying biological mechanism.

Cross-dataset transfer. Applied without re-training, AUROC on Herlev is 0.728 and balanced accuracy on APACC and HiCervix is 0.535 and 0.562 (Table 1; Herlev confusion matrix in Figure 10). Performance therefore

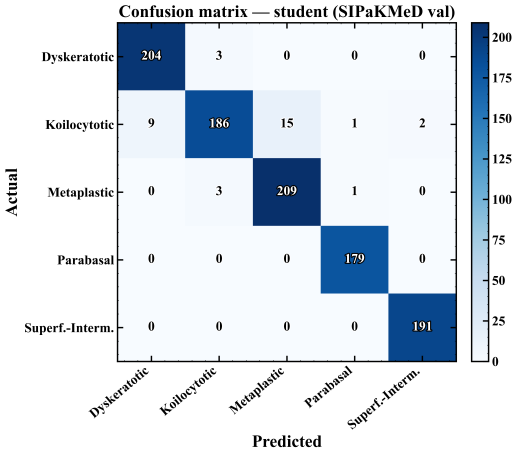


Figure 7. Student five-class confusion matrix (SIPaKMeD validation split).

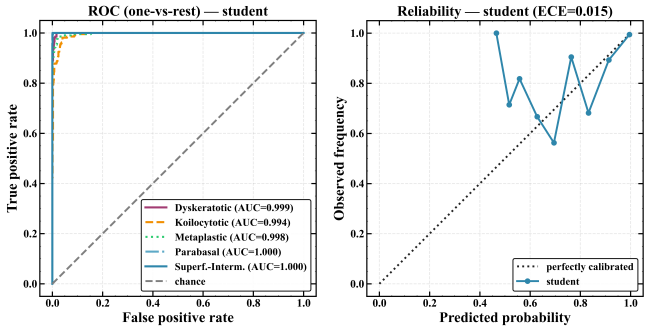


Figure 8. Student ROC curves (left) and reliability diagram (right; ECE = 0.016).

varies substantially across collections, indicating marked sensitivity to acquisition and population differences rather than uniform generalisation.

5.2 HPV-risk expert

Model comparison. Among the five families (Figure 11, Table 2) the tree ensembles lead on AUROC; RandomForest is best, with 0.688 AUROC and 0.721 accuracy (0.397 F1, 0.423 precision, 0.374 recall) under 5-fold cross-validation on the real cohort. Discrimination from questionnaire data alone is modest but consistent across the tree families (AUROC 0.68–0.69).

Best-model evaluation. For RandomForest, the confusion matrix, ROC, and precision–recall curves (Figure 12) give AUROC 0.688; at the default 0.5 threshold precision is 0.423 and recall 0.374. From questionnaire inputs alone the model yields a modest but usable pre-test risk signal—helpful for prioritising who to test rather than as a standalone diagnostic—and the operating point can be shifted

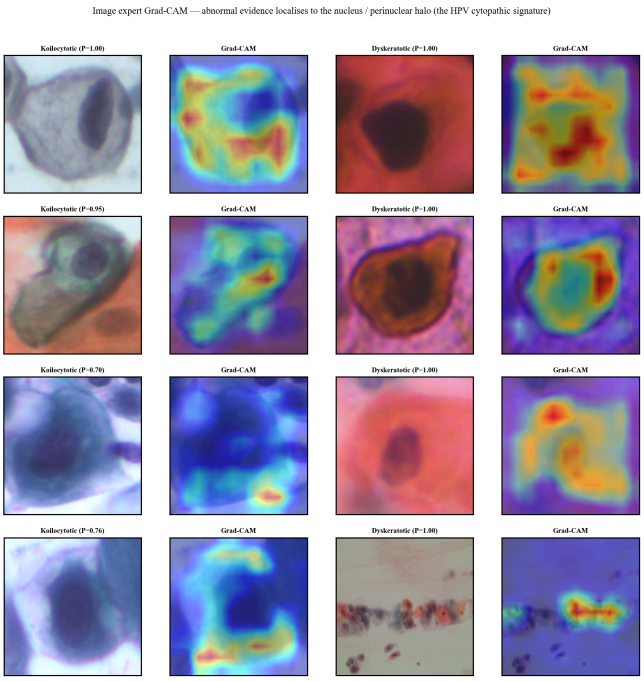


Figure 9. Grad-CAM on high-confidence Koilocytotic and Dyskeratotic cells: attribution concentrates on nuclear/perinuclear regions.

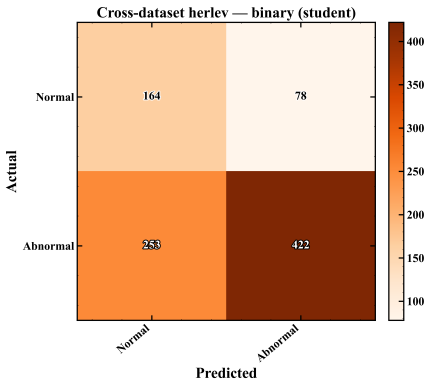


Figure 10. Binary (normal/abnormal) confusion matrix on the external Herlev dataset (AUROC 0.728).

along the ROC/PR curves toward higher recall when used as a triage flag.

Calibration and drivers. After isotonic calibration the reliability curve follows the diagonal (Figure 13). Feature importance (Figure 14) and SHAP attribution [9] (Figure 15) agree that behavioural-exposure variables—marital status, lifetime number of partners, age, and the partner-acquisition rate—have the largest effect, in the directions expected from HPV epidemiology, which supports that the model captures genuine risk structure rather than dataset artefacts.

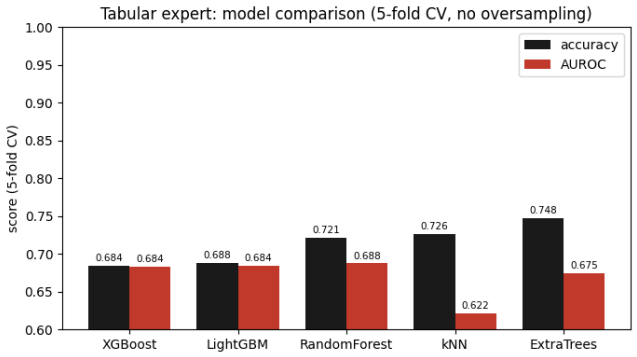


Figure 11. HPV-risk model comparison (5-fold CV): accuracy and AUROC per classifier family.

Table 2. HPV-risk expert on NHANES (n = 12,634), 5-fold CV (class weights, no oversampling). RandomForest obtains the highest AUROC.

Model	Acc	AUROC	F1	Prec	Rec
RandomForest	0.721	**0.688**	0.397	0.423	0.374
LightGBM	0.688	0.684	0.439	0.394	0.497
XGBoost	0.684	0.684	0.444	0.391	0.514
ExtraTrees	0.748	0.675	0.236	0.460	0.159
k-NN	0.726	0.622	0.289	0.399	0.227

Table 3. Operating modes. The system runs whichever modality is present and returns the corresponding calibrated output.

Available input	Output	AUROC
image only	p_{img} (cytology)	0.998
questionnaire only	p_{tab} (HPV-risk)	0.688
both	p_{fused} (calibrated)	0.993

5.3 Fused system

Discrimination is not the point. On a constructed deployment cohort the cytology expert reaches 0.998 AUROC and the HPV-risk expert 0.70 (its 5-fold value is 0.688), and the fused output reaches 0.993 (Figures 16–17)—essentially matching the stronger single expert. The fusion therefore does *not* increase discrimination over the stronger single expert—nor is it expected to. Its purpose is to add complementary HPV-risk information, remain calibrated across deployment prevalences, and operate under a missing modality (Table 3).

***Calibration across prevalence (CO-PRIOR).*.** Without re-anchoring, calibration error grows as the deployment prevalence τ departs from the training prevalence; with CO-PRIOR it is reduced substantially across the tested

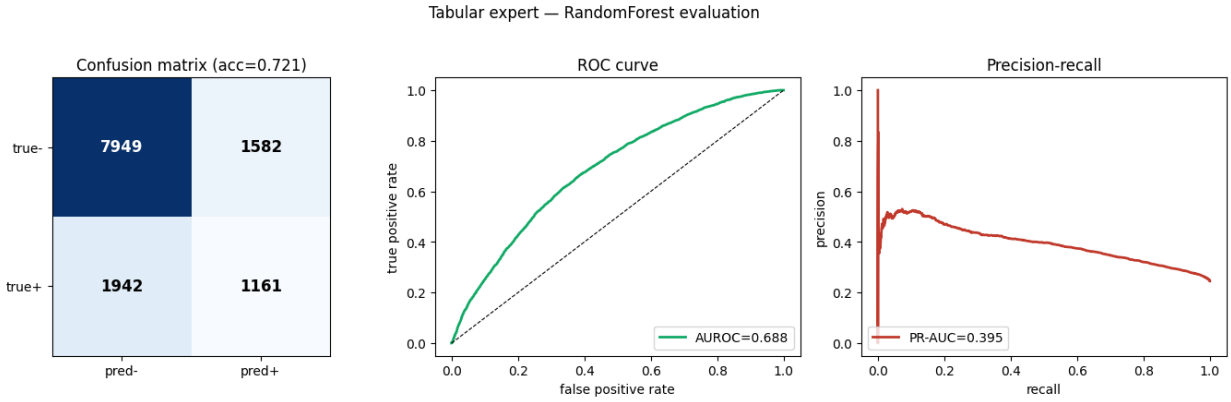


Figure 12. HPV-risk expert (RandomForest): confusion matrix (left), ROC (centre), precision–recall (right).

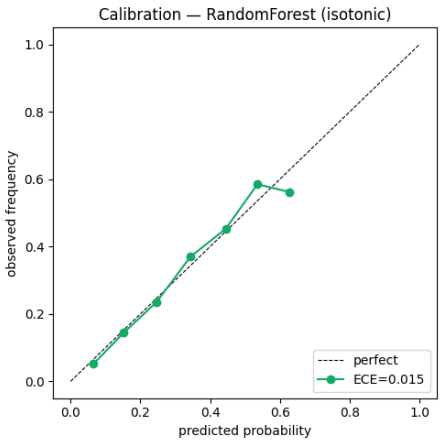


Figure 13. HPV-risk expert reliability diagram (isotonic-calibrated).

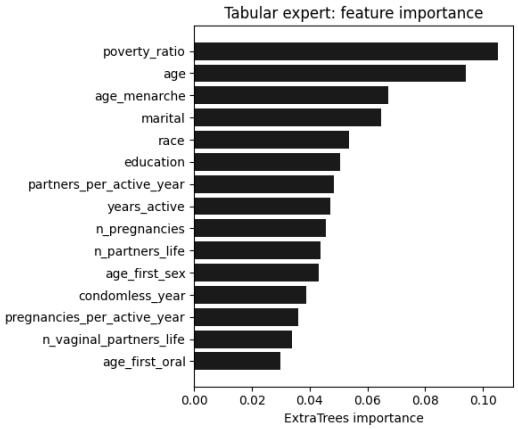


Figure 14. HPV-risk expert feature importance (impurity-based, tree ensemble).

range—for example ECE 0.21 → 0.00 at $\tau = 0.02$ (Figure 18). This is the main quantitative benefit of the fusion machinery.

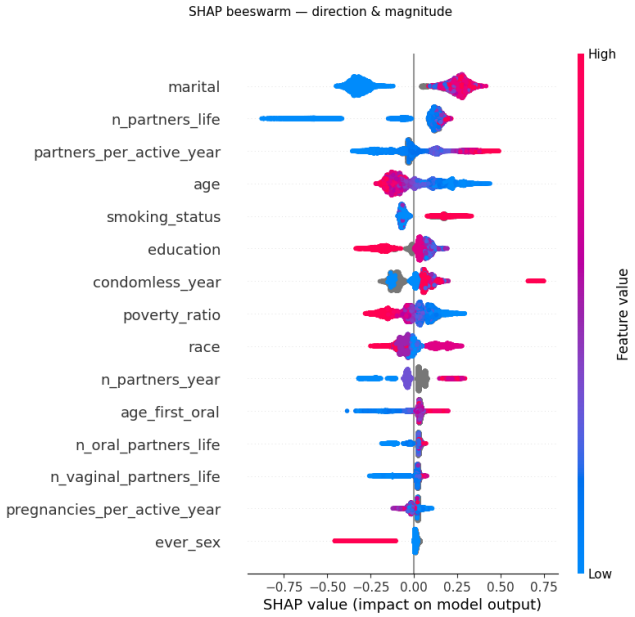


Figure 15. SHAP beeswarm for the HPV-risk expert (feature value in colour; horizontal position is the contribution toward the positive class).

Abstention coverage. Split-conformal coverage is close to the target for the HPV-risk expert (0.904 at target 0.90) and somewhat below target for the cytology expert (0.855 at target 0.90; Figure 19, Table 4). The coverage guarantee is therefore approximately met for the HPV-risk expert and close for the cytology expert at this level.

6 Discussion

The cytology expert performs strongly on its own endpoint (0.998 validation AUROC for cytological abnormality) and the HPV-risk expert gives a modest but usable signal (0.688 cross-validated AUROC for high-risk HPV)—and,

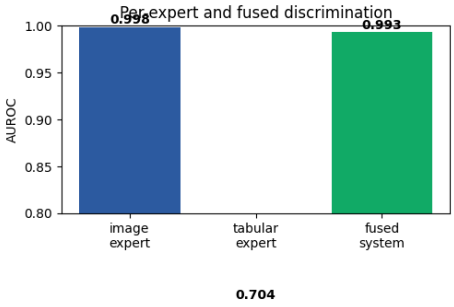


Figure 16. AUROC of each single expert and of the fused output on the constructed cohort.

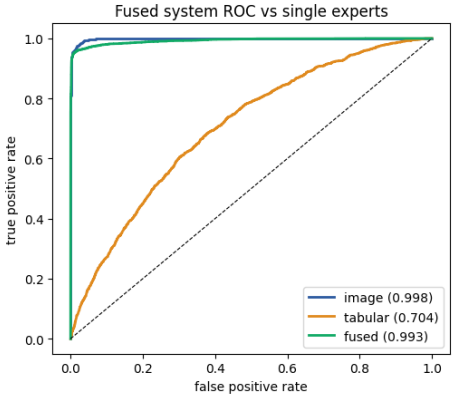


Figure 17. ROC curves of the cytology expert, HPV-risk expert, and fused output.

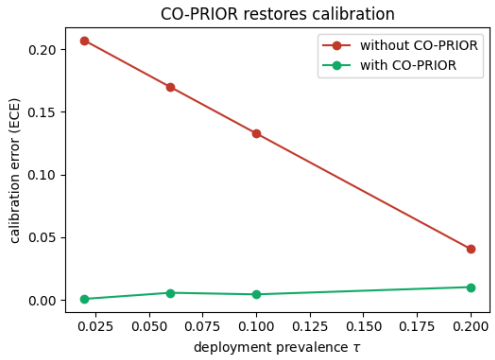


Figure 18. CO-PRIOR: expected calibration error versus deployment prevalence τ, with and without prior re-anchoring.

importantly, each emits a calibrated probability. The fusion does not improve discrimination beyond the stronger expert, and we do not claim it does; its role is to (i) reconcile the two differently-calibrated outputs onto one deployment-prevalence scale (CO-PRIOR), (ii) keep the fused probability calibrated when the local disease rate

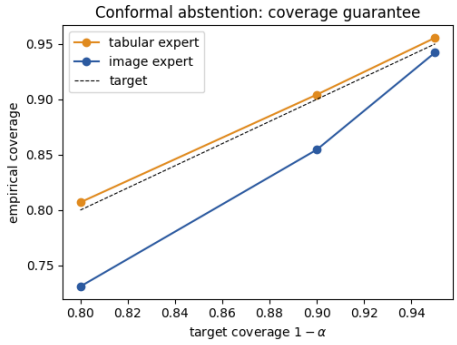


Figure 19. Split-conformal empirical coverage versus target $1-\alpha$ for each expert.

Table 4. Fused-system summary. Fusion does not raise discrimination; its measured benefit is calibration across prevalence.

Quantity	Value
Cytology expert AUROC	0.998
HPV-risk expert AUROC	0.688
Fused output AUROC	0.993
Fused ECE	0.043
CO-PRIOR ECE at $\tau = 0.02$	$0.21 \rightarrow 0.00$
Conformal coverage at target 0.90	0.904 (HPV-risk) / 0.855 (cytology)

differs from the training population, and (iii) return the appropriate calibrated output when only one modality is available. Framed this way, the value of CytoBridge is operational—combining heterogeneous, unpaired evidence with calibrated, missing-modality behaviour—rather than a discrimination gain.

7 Limitations

- *No independent test set for the cytology expert.* The 0.966/0.998 figures are validation-split, in-distribution numbers; a held-out or prospective test set would give a less optimistic estimate.
- *Unpaired modalities.* No patient has both an image and a questionnaire, so the fused decision is analysed per expert and on a constructed cohort, not measured on real paired data.
- *Domain shift.* External image performance drops markedly (Herlev 0.728; APACC 0.535; HiCervix 0.562 balanced accuracy), so in-distribution results do not transfer uniformly.

8 Conclusion

We asked whether independently trained cytology and questionnaire HPV-risk models can be combined at the decision level to give calibrated, any-modality screening support without paired patient data. The two experts reach

0.998 (validation) and 0.688 (cross-validation) AUROC on their respective endpoints. Combining them with CO-PRIOR prior re-anchoring, an asymmetric combiner, and conformal abstention does not increase discrimination beyond the stronger expert, but it keeps the fused probability calibrated across deployment prevalences and lets the system operate when a modality is missing. The main open problems—an independent and ideally paired test cohort, and the observed domain-shift sensitivity of the image model—are the natural next steps toward clinical evaluation.

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